

2015 GCSE Syllabus Worksheets

Foundation Worksheets

Grades 1 to 5

Grade 1

Grade 2

Grade 3

Grade 4

Grade 5

Part 1

Instructions for use

Clicking on something like this **Grade 4** will take you to the Grade 4 clip numbers and titles.


Clicking on something like this Compound Units. 142 will take you to Worksheet 142.


Clicking on something like this

©MathsWatch	Clip 142
Grade 4 questions	

 will take you back to the contents page.


Take me to . . .

Grade 1

Grade 2

Grade 3

Grade 4

Grade 5

Grade 3 Foundation and Higher

Multiplying Decimals	66
Dividing Decimals.	67
Four Rules of Negatives.	68
Listing Strategies	69
Comparing Fractions	70
Adding and Subtracting Fractions	71
Finding a Fraction of an Amount	72
Multiplying Fractions	73
Dividing Fractions.	74
BODMAS/BIDMAS.	75
Reciprocals	76
Calculator Questions	77
Product of Primes.	78
Highest Common Factor (HCF)	79
Lowest Common Multiple (LCM)	80 A/B
Squares, Cubes and Roots	81
Working with Indices	82
Standard Form	83
Decimals and Fractions	84
Fractions, Percentages, Decimals	85
Percentage of an Amount (Calc.)	86
Percentage of an Amount (Non-Calc.)	87
Change to a Percentage (Calc.)	88
Change to a Percentage (Non-Calc.)	89 A/B
Rounding to Significant Figures	90
Estimating Answers	91
Using Place Value	92

1) Work out

a) 6×0.2

b) 0.2×0.3

c) 0.4×7

d) 0.2×0.8

e) 0.03×0.9

f) 1.5×0.2

2) A box contains 7 books, each weighing 2.5 kg.

Work out the total weight of the box.

3) Jim takes 13 boxes out of his van.

The weight of each box is 25.5 kg.

Work out the total weight of the 13 boxes.

4) Tim has a job which pays £6.85 per hour.

If he works for 34 hours, one week, how much does he earn?

5) Sue has a part-time job and the hourly pay is £7.50 per hour.

How much does she earn if she works for 8.5 hours, one week?

6) Fencing costs £13.25 per metre.

How much does 12.5 m cost?

1) Work out

a) $9 \div 0.3$

b) $6 \div 0.1$

c) $12 \div 0.4$

d) $25 \div 0.5$

e) $21 \div 0.3$

f) $15 \div 0.2$

2) Work out

a) $3.6 \div 0.4$

b) $0.8 \div 0.2$

c) $2.4 \div 0.4$

d) $0.56 \div 0.08$

e) $5.5 \div 0.05$

f) $8.1 \div 0.09$

3) John takes boxes out of his van.

The total weight of the boxes is 4.9 kg

The weight of each box is 0.7 kg

Work out the number of boxes in John's van.

4) Mr Rogers bought a bag of elastic bands for £6

Each elastic band costs 12p.

Work out the number of elastic bands in the bag.

1) Work out the following:

a) $2 - 7$

b) $4 - 6$

c) $1 - 8$

d) $0 - 4$

2) Work out the following:

a) $-3 + 2$

b) $-7 + 5$

c) $-3 + 8$

d) $-9 + 11$

3) Work out the following:

a) $-1 - 3$

b) $-4 - 5$

c) $-7 - 8$

d) $-2 - 12$

4) Work out the following:

a) $6 - -3$

b) $-3 - -5$

c) $-9 - -2$

d) $1 - -13$

5) Work out the following:

a) -3×4

b) 5×-2

c) -4×-5

d) -6×-3

6) Work out the following:

a) $12 \div -4$

b) $-20 \div -2$

c) $-15 \div 3$

d) $-100 \div -5$

Listing Strategies

- 1) A restaurant has the simple menu, as shown.

Meg chooses a starter, a main course and a dessert.

List all the possible combinations that Meg could choose.

<u>Starter</u>
Soup
Melon
<u>Main Course</u>
Chicken
Steak
Vegetarian
<u>Dessert</u>
Ice Cream
Trifle

- 2)



- Make a list of all the two-digit numbers that can be made with these three cards.
- Make a list of all the three-digit numbers that can be made with these three cards.

- 3) Phil has three different coins.

He has: a £2 coin

a £1 coin

a 50p coin

- If he chooses exactly two coins, what amounts of money can he make?
- If he chooses one or two coins, what amounts of money can he make?
- If he chooses one, two or three coins, what amounts of money can he make?

Comparing Fractions

- 1) Put these fractions in order of size, smallest to largest.
Show your working for each question.

a) $\frac{1}{2}$ $\frac{1}{3}$

b) $\frac{3}{5}$ $\frac{2}{3}$

c) $\frac{1}{2}$ $\frac{3}{8}$

- 2) Put these fractions in order of size, smallest to largest.
Show your working for each question.

a) $\frac{1}{2}$ $\frac{1}{4}$ $\frac{3}{8}$

b) $\frac{3}{5}$ $\frac{1}{2}$ $\frac{3}{4}$

c) $\frac{5}{6}$ $\frac{2}{3}$ $\frac{3}{4}$

- 3) Put these fractions in order of size, smallest to largest.
Show your working for each question.

a) $\frac{2}{3}$ $\frac{7}{12}$ $\frac{3}{4}$ $\frac{5}{6}$

b) $\frac{5}{8}$ $\frac{2}{3}$ $\frac{3}{24}$ $\frac{7}{12}$

c) $\frac{6}{10}$ $\frac{4}{5}$ $\frac{5}{12}$ $\frac{8}{15}$

- 4) Ben spent his pocket money this way:

$\frac{7}{20}$ on magazines

$\frac{4}{10}$ on chocolates

$\frac{1}{4}$ on games

Order the items Ben bought by value, largest first.
Show all your working.

Adding and Subtracting Fractions

In all the questions on this page, please give your answers in their simplest form.

1) Work out the following:

a) $\frac{1}{7} + \frac{3}{7}$

b) $\frac{4}{9} + \frac{1}{9}$

2) Work out the following:

a) $\frac{1}{5} + \frac{3}{4}$

b) $\frac{3}{8} + \frac{1}{4}$

c) $\frac{2}{3} + \frac{3}{10}$

d) $\frac{1}{2} + \frac{2}{5}$

3) Work out the following:

a) $\frac{2}{3} + \frac{1}{2}$

b) $\frac{3}{5} + \frac{2}{3}$

c) $\frac{5}{8} + \frac{3}{4}$

d) $\frac{5}{7} + \frac{2}{5}$

4) Work out the following:

a) $2\frac{1}{2} + 1\frac{3}{4}$

b) $1\frac{2}{5} + \frac{2}{3}$

c) $2\frac{1}{6} + 1\frac{1}{2}$

d) $1\frac{3}{7} + \frac{2}{5}$

5) Work out the following:

a) $\frac{3}{4} - \frac{1}{2}$

b) $\frac{5}{7} - \frac{2}{3}$

c) $\frac{5}{8} - \frac{1}{3}$

d) $\frac{8}{9} - \frac{2}{3}$

6) Work out the following:

a) $2\frac{1}{2} - 1\frac{3}{4}$

b) $1\frac{2}{3} - \frac{3}{4}$

c) $3\frac{2}{5} - 1\frac{1}{2}$

d) $2\frac{3}{8} - \frac{3}{5}$

7) Ted received his pocket money on Friday.

He spent $\frac{3}{5}$ of his pocket money on games.

He spent $\frac{1}{10}$ of his pocket money on magazines.

What fraction of his pocket money did he have left?

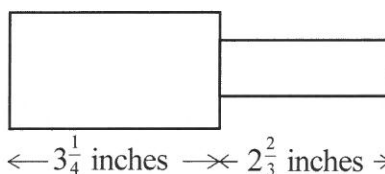
8) Maisie buys a bag of flour.

She uses $\frac{1}{4}$ to bake a cake and $\frac{2}{5}$ to make a loaf.

a) What fraction of the bag of flour was used?

b) What fraction of the bag of flour is left?

9) Work out the total length of this shape.
Give your answer as a mixed number.



Finding a Fraction of an Amount

1) Work out these amounts.

a) $\frac{3}{4}$ of £20

b) $\frac{2}{3}$ of 60 kg

c) $\frac{3}{8} \times 24$

d) $150 \times \frac{2}{3}$

e) $\frac{2}{9}$ of 180 cm

f) $49 \times \frac{4}{7}$

g) $60 \times \frac{1}{4}$

h) $\frac{5}{8}$ of £48

i) $4000 \times \frac{7}{8}$

2) There are 600 apples on a tree and there are maggots in $\frac{3}{5}$ of them.

How many apples have maggots in them?

3) Liz and Lee are travelling in a car from Glasgow to Poole (770 km).

At midday they had already travelled $\frac{5}{7}$ of the total distance.

What distance, in km, had they travelled by midday?

4) A digital camera that cost £49 was sold on eBay for $\frac{3}{7}$ of the original price.

What was the selling price?

5) Yesterday Thomas travelled a total of 175 miles.

He travelled $\frac{2}{5}$ of this distance in the morning.

How many miles did he travel during the rest of the day?

6) Debra received her £15 pocket money on Saturday.

She spent $\frac{1}{3}$ of her pocket money on magazines.

She spent $\frac{2}{5}$ of her pocket money on a necklace.

How much of the £15 did she have left?

In all the questions on this page, please give your answers in their simplest form.

1) Work out the following:

a) $\frac{1}{2} \times \frac{1}{2}$

b) $\frac{2}{3} \times \frac{1}{3}$

c) $\frac{3}{5} \times \frac{2}{7}$

d) $\frac{4}{7} \times \frac{5}{9}$

2) Work out the following:

a) $\frac{1}{2} \times \frac{2}{3}$

b) $\frac{3}{4} \times \frac{8}{11}$

c) $\frac{2}{9} \times \frac{3}{4}$

d) $\frac{4}{5} \times \frac{1}{12}$

3) Work out the following:

a) $1\frac{1}{2} \times \frac{1}{3}$

b) $\frac{2}{3} \times 2\frac{2}{5}$

c) $3\frac{1}{2} \times 1\frac{1}{2}$

d) $1\frac{2}{7} \times 3\frac{1}{3}$

In all the questions on this page, please give your answers in their simplest form.

1) Work out the following:

a) $\frac{2}{5} \div \frac{3}{4}$

b) $\frac{1}{7} \div \frac{3}{5}$

c) $\frac{4}{9} \div \frac{1}{2}$

d) $\frac{3}{10} \div \frac{5}{9}$

2) Work out the following:

a) $\frac{1}{2} \div \frac{1}{3}$

b) $\frac{3}{7} \div \frac{4}{7}$

c) $\frac{1}{9} \div \frac{2}{3}$

d) $\frac{2}{5} \div \frac{3}{10}$

3) Work out the following:

a) $1\frac{1}{3} \div \frac{1}{4}$

b) $\frac{3}{5} \div 2\frac{2}{3}$

c) $3\frac{2}{3} \div 1\frac{1}{5}$

d) $4\frac{1}{2} \div 1\frac{1}{2}$

1) $6 \times 5 + 2$

2) $2 + 6 \times 5$

3) $35 - 4 \times 3$

4) $48 \div (14 - 2)$

5) $27 \div (3 + 6)$

6) $27 \div 3 + 6$

7) $(9 + 2) \times 2 + 5$

8) $4 \times (1 + 4) - 6$

9) $6 \times 4 - 3 \times 5$

10) $\frac{9+3}{4+2}$

11) $\frac{23+9}{7-3}$

12) $\frac{7-2^2}{4^2-15}$

13) $\frac{5^2+3}{2 \times 7}$

14) $\frac{5 \times 6 - 4}{13}$

15) $\frac{8 \times 2 - 4}{3 + 1^2}$

16) $\frac{12 - 3 \times 2}{14 \div 7}$

17) $\frac{20 - 3^2}{10 - (5 + 4)}$

18) $\frac{3 + 9 \times 8}{1 + 6 \times 4}$

1) Write down the reciprocal of

a) 8

b) 3

c) 1

d) 12

2) Write down the reciprocal of

a) $\frac{1}{2}$

b) $\frac{1}{3}$

c) $\frac{4}{3}$

d) $\frac{5}{8}$

3) Write down the reciprocal of

a) 0.1

b) 0.5

c) 0.2

4) Why can't we have a reciprocal of 0?



- 1) Use your calculator to work out

$$\frac{23.7 \times 14.2}{8.4 \times 3.2}$$

Write down all the figures on your calculator display.



- 2) Use your calculator to work out

$$\frac{\sqrt{21.4}}{5.7 - 2.35}$$

Write down all the figures on your calculator display.



- 3) Work out $\frac{5.8 + 4.65}{3.1^2 + 1.62}$

Write down all the figures on your calculator display.



- 4) Use your calculator to work out the value of

$$\frac{9.2 \times 16.3}{9.4 - 5.71}$$

Write down all the digits from your calculator.
Give your answer as a decimal.



- 5) Use your calculator to work out

$$\frac{3}{2.1 + 3.45}$$

Write down all the figures on your calculator display.
You must give your answer as a decimal.



- 6) Use your calculator to work out

$$\frac{15^2 - 12^2}{\sqrt{9.6 - 3.87}}$$

Write down all the figures on your calculator display.
You must give your answer as a decimal.



- 7) Use a calculator to work out

$$\sqrt{\frac{22.4 \times 13.9}{3.6}}$$

Write down all the figures on your calculator display.

- 1) List the first seven prime numbers.
- 2) Express the following numbers as the product of their prime factors:
 - a) 12
 - b) 20
 - c) 30
 - d) 24
- 3) Express the following numbers as the product of their prime factors:
 - a) 64
 - b) 100
 - c) 150
- 4) Express the following numbers as the product of their prime factors:
 - a) 175
 - b) 192
 - c) 315
- 5) The number 96 can be written as $2^m \times n$, where m and n are prime numbers.
Find the value of m and the value of n .
- 6) The number 75 can be written as $5^x \times y$, where x and y are prime numbers.
Find the value of x and the value of y .

Highest Common Factor (HCF)

- 1) Find the Highest Common Factor of 16 and 24.

- 2) Find the Highest Common Factor of 21 and 28.

- 3) Find the Highest Common Factor of 60 and 150.

- 4) Find the Highest Common Factor of 96 and 108.

- 5)
 - (i) Write 42 and 63 as products of their prime factors.
 - (ii) Work out the Highest Common Factor of 42 and 63.

Lowest Common Multiple (LCM)

- 1) Find the Lowest Common Multiple of 20 and 60.

- 2) Find the Lowest Common Multiple of 28 and 72.

- 3) Find the Lowest Common Multiple of 70 and 240.

- 4) Find the Lowest Common Multiple of 35 and 55.

- 5)
 - (i) Write 42 and 63 as products of their prime factors.
 - (ii) Work out the Lowest Common Multiple of 42 and 63.

- 1) a) Express 84 as a product of its prime factors.
b) Find the Highest Common Factor (HCF) of 84 and 35.

2) Express 72 as the product of its prime factors.

3) Express 180 as the product of its prime factors.

- 4) a) Express 66 as a product of its prime factors.
b) Express 132^2 as a product of its prime factors.



5) Express 252 as a product of its prime factors.



6) Find the Lowest Common Multiple (LCM) of 24 and 36.



- 7) a) Write 56 as a product of its prime factors.
b) Find the Highest Common Factor (HCF) of 56 and 42.



- 8) a) Express 45 as a product of its prime factors.
b) Find the Highest Common Factor (HCF) of 45 and 30.



- 9) a) Find the Highest Common Factor (HCF) of 24 and 30.
b) Find the Lowest Common Multiple (LCM) of 4, 5 and 6.

- 1) What is the value of 5^2 ?
- 2) What is the value of 8^2 ?
- 3) These are the first five square numbers: 1, 4, 9, 16, 25
 - a) What is the sixth square number?
 - b) What is the 10th square number?
- 4) Which square number lies between 60 and 70?
- 5) What is the value of 2^3 ?
- 6) What is the value of 4^3 ?
- 7) Work out $1^3 + 2^3 + 3^3$
- 8) Work out $\sqrt{25}$
- 9) Work out $\sqrt{49}$
- 10) Work out the value of $\sqrt{121} \times \sqrt{121}$
- 11) Match together cards with the same answer

9^2	$\sqrt{9}$	81	5^3
2^5	125	32	3

1) Evaluate the following:

a) 2^3

b) 3^2

c) 10^4

2) Evaluate the following:

a) 2^8

b) 6^4

c) 5^6

3) Find the value of

a) $2^4 + 3^2$

b) $5^2 - 2^3$

c) $1^2 + 2^2 + 3^2$

4) Find the value of

a) $5^4 + 6^3$

b) $3^4 \times 2^5$

c) $9^3 - 6^3$

5) Find the value of

$$2^2 + 3^2 + 5^2 + 7^2 + 11^2 + 13^2 + 17^2$$

1) Change the following to normal (or ordinary) numbers.

a) 4.3×10^4

c) 7.03×10^3

e) 1.01×10^4

b) 6.79×10^6

d) 9.2×10^2

f) 4×10^5

2) Change the following to normal (or ordinary) numbers.

a) 4.3×10^{-4}

c) 7.03×10^{-3}

e) 1.01×10^{-4}

b) 6.79×10^{-6}

d) 9.2×10^{-2}

f) 4×10^{-5}

3) Change the following to standard form.

a) 360

c) 520 000

e) 1 003

b) 8 900

d) 60000

f) 6 450 000

4) Change the following to standard form.

a) 0.071

c) 0.00076

e) 0.00009

b) 0.0008

d) 0.0928

f) 0.00000173

5) Work out the following, giving your answer in standard form.

a) $3\,000 \times 5\,000$

d) $5 \times 4 \times 10^3$

g) $7 \times 10^2 \times 3 \times 10^{-4}$

b) 240×0.0002

e) $\frac{8 \times 10^4}{4 \times 10^2}$

h) $2 \times 3.6 \times 10^{-5}$

c) $9 \times 1.1 \times 10^7$

f) $9 \times 10^2 \times 2 \times 10^{-5}$

i) $6 \times 4.1 \times 10^3$

Write the following fractions as decimals

1) $\frac{3}{10}$

2) $\frac{7}{10}$

3) $\frac{9}{100}$

4) $\frac{1}{2}$

5) $\frac{3}{4}$

6) $\frac{2}{5}$

7) $\frac{7}{20}$

8) $\frac{1}{3}$

9) $\frac{1}{8}$

10) $\frac{5}{8}$

1) Write the following fractions as decimals and percentages:

eg. $\frac{1}{10} \xrightarrow{1 \div 10} 0.1 \xrightarrow{0.1 \times 100} 10\%$

a) $\frac{3}{10} =$

b) $\frac{1}{5} =$

c) $\frac{2}{5} =$

d) $\frac{1}{4} =$

e) $\frac{3}{4} =$

f) $\frac{1}{2} =$

g) $\frac{1}{3} =$

2) Fill in the blanks in the table below:

Fraction	Decimal	Percentage
$\frac{6}{10}$		
	0.2	
	0.9	
		40%
		25%
$\frac{4}{5}$		
$\frac{12}{100}$		
	$0.\dot{3}$	
		70%



1) Work out:

- a) 21% of £340
- b) 64% of £1080
- c) 36% of £800
- d) 98% of £13



2) Work out:

- a) 17.5% of £58
- b) 20% of £5.40
- c) 61.7% of £2000
- d) 17.5% of £68.40



3) A computer costs £406 plus VAT at 20%.
Work out the total cost of the computer.



4) A car is usually priced at £9800 but now has a discount of 8%.
What is the new price of the car?



5) 9500 people attend a festival and 22% of them are children.
How many children are at the festival?



6) 65% of a car, by weight, is steel and iron.
If a car weighs 1100 kg, what is the weight of steel and iron in the car?



7) Tony earns £17800 per year and receives a 3.8% pay rise.
How much does he now earn?

- 1) Work out:
 - a) 10% of £170
 - b) 10% of £6800
 - c) 10% of £923
 - d) 10% of £16

- 2) Work out:
 - a) 20% of £60
 - b) 30% of £90
 - c) 15% of £800
 - d) 15% of £68

- 3) Work out:
 - a) 35% of £80
 - b) 90% of £160
 - c) 17.5% of £600
 - d) 17.5% of £850

- 4) Work out:
 - a) 15% of £4.60
 - b) 40% of £2.80
 - c) 17.5% of £3.20
 - d) 97.5% of £24

- 5) The normal price of a jacket is £54.
In a sale, the price is reduced by 30%
What is the sale price?

- 6) A football costs £14 plus 20% VAT.
How much is the football?



1) Write the following as percentages, giving all your answers to 1 decimal place.

- a) 12 out of 34
- b) 62 out of 85
- c) 113 out of 153
- d) 2150 out of 3452



2) Sarah sat a Science test and got a score of 64 marks out of 112 possible marks.
What was her mark as a percentage?
Give your answer to 1 decimal place.



3) In a class of 32 students, 18 of them are boys.
What percentage of the class are boys?
Give your answer to 1 decimal place.



4) In a French class there are 13 girls and 6 boys.
What percentage of the class are girls?
Give your answer to 1 decimal place.



5) A new car usually costs £8500.
Henry gets a discount of £1000.
What is the discount as a percentage of the usual cost?
Give your answer to 1 decimal place.



6) Write out £148 as a percentage of £600.
Give your answer to 1 decimal place.



7) In a wood there are 200 oak trees, 650 silver birch trees and 400 wild cherry trees.
What percentage of the trees are oak trees?



8) In England in 2010 there were 68820 deaths caused by cancer.
Of these deaths, 37500 were caused by smoking.
What percentage of deaths due to cancer were caused by smoking?
Give your answer to 1 decimal place.

- 1) Write the following as percentages.
 - a) 12 out of 50
 - b) 15 out of 25
 - c) 8 out of 10
 - d) 11 out of 20
 - e) 4 out of 5
 - f) 32 out of 40
 - g) 12 out of 80
 - h) 640 out of 800
 - i) 36 out of 60

- 2) Tim got 17 out of 20 in a French test.
Write 17 out of 20 as a percentage.

- 3) Write £19 as a percentage of £25

- 4) Work out £14 as a percentage of £40

- 5) A baker burnt 12 loaves out of the 200 loaves he baked.
What percentage of the 200 loaves did he burn?

- 6) What is £380 as a percentage of £400?

- 7) What is £22 as a percentage of £40?

- 8) If there are 9 girls and 11 boys in a class, what percentage of the class are girls?

Percentages



- 1) Find the following to the nearest penny:
- 23% of £670
 - 12% of £580
 - 48% of £64
 - 13% of £7.50
 - 87% of £44
 - 15.7% of £7000
 - 23.8% of £980
 - 34% of £16.34
 - 48.6% of £971.26
 - 78.24% of £12.82
 - 42.15% of £7876.42
 - 0.57% of £60000

2) Find the following:

- 10% of £700
- 10% of £400
- 10% of £350
- 10% of £530
- 10% of £68
- 10% of £46
- 10% of £6.50
- 10% of £12.20
- 20% of £600
- 30% of £900
- 60% of £800
- 20% of £650
- 40% of £320
- 15% of £300
- 15% of £360
- 65% of £12000
- 45% of £64
- 85% of £96
- 17.5% of £800
- 17.5% of £40
- 17.5% of £8.80



3) Change the following to percentages, giving all answers to 1 decimal place:

- 6 out of 28
- 18 out of 37
- 42 out of 83
- 24 out of 96
- 73 out of 403
- 234 out of 659
- 871 out of 903
- 4.7 out of 23
- 6.9 out of 79
- 14.8 out of 23.6
- 65.8 out of 203.7

4) Change the following to percentages:

- 46 out of 100
- 18 out of 50
- 7 out of 25
- 23 out of 25
- 9 out of 20
- 16 out of 20
- 7 out of 10
- 9.5 out of 10
- 10 out of 40
- 16 out of 40
- 30 out of 40
- 12 out of 40
- 28 out of 80
- 32 out of 80
- 60 out of 80
- 3 out of 5
- 4 out of 5
- 15 out of 75
- 24 out of 75
- 30 out of 75

- 5) A shop gives a discount of 20% on a magazine that usually sells for £2.80. Work out the discount in pence.



- 6) A television costs £596 plus VAT at 17.5%.
Work out the cost of the television including VAT.



- 7) Peter has 128 trees in his garden. 16 of the trees are pear trees.
What percentage of the trees in his garden are pear trees?



- 8) Jane scored 27 out of 42 in a Maths test and 39 out of 61 in a Science test.
What were her percentages in both subjects to 1 decimal place?

- 9) In class 9A there are 7 girls and 18 boys.
What percentage of the class are girls?

- 10) A shop decides to reduce all the prices by 15%.
The original price of a pair of trainers was £70. How much are they after the reduction?



- 11) VAT at 17.5% is added to the price of a car. Before the VAT is added it cost £18000.
How much does it cost with the VAT?

Rounding to Significant Figures

1) Round the following numbers to 1 significant figure:

- a) 428
- b) 783
- c) 5608
- d) 3 521
- e) 21 999
- f) 793 041

2) Round the following numbers to 2 significant figures:

- a) 846
- b) 2 647
- c) 3 552
- d) 46 817
- e) 89 711
- f) 195 084

3) Round the following numbers to 3 significant figures:

- a) 91 249
- b) 64 182
- c) 223 058
- d) 389 512
- e) 7 761 223
- f) 4 997 124



4) Work out the following and give your answer to 3 significant figures:

- a) 216×348
- b) 7721×609
- c) 8714×2198

5) Round the following numbers to 1 significant figure:

- a) 0.00618
- b) 0.00482
- c) 0.00006492
- d) 0.004981

6) Round the following numbers to 2 significant figures:

- a) 0.035812
- b) 0.00082477
- c) 0.0038611
- d) 0.000037211

7) Round the following numbers to 3 significant figures:

- a) 0.00143229
- b) 0.000721981
- c) 0.0000044251
- d) 0.000668821

8) Round the following numbers to 3 significant figures:

- a) 47.84122
- b) 9.778112
- c) 12.35913



9) Work out the following and give your answer to 3 significant figures:

- a) $15 \div 0.38$
- b) $0.31 \div 0.16$
- c) 208×366

1) Work out an estimate for $\frac{304 \times 9.96}{0.51}$

2) Work out an estimate for $\frac{6.7 \times 192}{0.051}$

3) Work out an estimate for $\frac{32 \times 4.92}{0.21}$

4) Work out an estimate for $\frac{3880}{236 \times 4.85}$

5) Work out an estimate for $\frac{7.18 \times 19.7}{0.47}$

1) Using the information that

$$4.7 \times 34 = 159.8$$

write down the value of

- a) 47×34
- b) 4.7×3.4
- c) $159.8 \div 47$

2) Using the information that

$$324 \times 48 = 15552$$

write down the value of

- a) 3.24×4.8
- b) 0.324×0.48
- c) $15552 \div 4.8$

3) Using the information that

$$73 \times 234 = 17082$$

write down the value of

- a) 730×234
- b) 73×2.34

4) Using the information that

$$27 \times 5.6 = 151.2$$

write down the value of

- a) 27×56
- b) 2.7×0.56
- c) $151.2 \div 56$

5) Using the information that

$$719 \times 35 = 25165$$

write down the value of

- a) 71.9×3.5
- b) 0.719×0.35
- c) $25165 \div 7.19$

